Assignment 6

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```
;Write x86 Assembly language program (ALP) to convert 4-digit HEX number into its
;equivalent BCD number and 5 digits BCD number into its equivalent HEX number. Make
;your program user friendly to accept the choice from user for
;a. Hex to BCD
;b. BCD to HEX
;c.EXIT.
%macro IO 4
    mov eax, %1
    mov ebx, %2
    mov ecx, %3
    mov edx, %4
    int 80h
%endmacro

section .data
    menu db 10, " MENU"
        db 10, "1. Hex to BCD"
        db 10, "2. BCD to Hex"
        db 10, "3. Exit"
        db 10, "Enter your choice: "
    menulen equ $ - menu

    msg1 db 10, "Enter Hex Number: "
    lenm1 equ $ - msg1

    msg2 db 10, "Enter BCD Number: "
    lenm2 equ $ - msg2

    msg3 db 10, "Equivalent BCD Number: "
    lenm3 equ $ - msg3
```

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    msg4 db 10, "Equivalent Hex Number: "
    lenm4 equ $ - msg4

    error db 10, "Invalid Choice!"
    lenerr equ $ - error

section .bss
    choice resb 1
    num resb 8
    answer resb 8
    factor resb 8

section .code
    global _start

_start:

MENU:
    IO 4, 1, menu, menulen
    IO 3, 0, choice, 2
    cmp byte [choice], '1'
    je hex2bcd
    cmp byte [choice], '2'
    je bcd2hex
    cmp byte [choice], '3'
    je exit
    IO 4, 1, error, lenerr
    jmp exit

    JMP MENU

hex2bcd:
```

```
    IO 4, 1, msg1, lenm1
    IO 3, 0, num, 9
    call asciihextohex

    mov eax, ebx
    mov ebx, 10
    mov edi, num + 7

loop3:
    mov edx, 0
    div ebx
    add dl, 30h
    mov [edi], dl
    dec edi
    cmp eax, 0
    jne loop3

    IO 4, 1, msg3, lenm3
    IO 4, 1, num, 8
RET

bcd2hex:
    IO 4, 1, msg2, lenm2
    IO 3, 0, num, 9

    mov ecx, 8
    mov esi, num + 7
    mov ebx, 0
    mov dword [factor], 1

loop4:
    mov eax, 0
```

```
    mov al, [esi]
    sub al, 30h
    mul dword [factor]
    add ebx, eax
    mov eax, 10
    mul dword [factor]
    mov dword [factor], eax
    dec esi
    loop loop4

    MOV EDI, EBX
    IO 4, 1, msg4, lenm4
    mov eax, EDI
    call display
    RET

exit:
    mov eax, 1
    mov ebx, 0
    int 80h

asciihextohex:
    mov esi, num
    mov ecx, 8
    mov ebx, 0
    mov eax, 0

loop1:
    rol ebx, 04
    mov al, [esi]
    cmp al, 39h
    jbe skip1
```

```
    sub al, 07h

skip1:
    sub al, 30h
    add ebx, eax
    inc esi
    dec ecx
    jnz loop1
ret

display:
    mov esi, answer + 7
    mov ecx, 8

loop2:
    mov edx, 0
    mov ebx, 16
    div ebx
    cmp dl, 09h
    jbe skip2
    add dl, 07h

skip2:
    add dl, 30h
    mov [esi], dl
    dec esi
    dec ecx
    jnz loop2

    IO 4, 1, answer, 8
ret
```

// OUTPUT

```
pllab0112@pllab0112-ThinkCentre-M70s: ~/Documents/nmiet/6th
pllab0112@pllab0112-ThinkCentre-M70s:~/Documents/nmiet/6th$ nasm -f elf mp4.asm
nasm: fatal: unable to open input file `mp4.asm' No such file or directory
pllab0112@pllab0112-ThinkCentre-M70s:~/Documents/nmiet/6th$ nasm -f elf conversion.asm
pllab0112@pllab0112-ThinkCentre-M70s:~/Documents/nmiet/6th$ ld -m elf_i386 -o conversion conversion.o
pllab0112@pllab0112-ThinkCentre-M70s:~/Documents/nmiet/6th$ ./conversion

 MENU
1. Hex to BCD
2. BCD to Hex
3. Exit
Enter your choice: 1

Enter Hex Number: 0000000F

Equivalent BCD Number: 00000015Segmentation fault (core dumped)
pllab0112@pllab0112-ThinkCentre-M70s:~/Documents/nmiet/6th$ ./conversion

 MENU
1. Hex to BCD
2. BCD to Hex
3. Exit
Enter your choice: 2

Enter BCD Number: 00000010

Equivalent Hex Number: 0000000ASegmentation fault (core dumped)
```